HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · RETAIL · E-COMMERCE

E-COMMERCE STUDIO OPERATIONS · PRODUCT PHOTOGRAPHY

Vision AI Image QC

Style guides became JSON rulebooks, cheap gatekeepers stopped junk images, and a vision model now judges every image set rule by rule, with evidence, before the set is struck.

~~48~~ → **5.1** QA hours per 1,000 images



90 % less
Manual image review time
▼ QA makes only the final call



26 %
Images stopped before the VLM
▼ fewer paid API calls



95 %
Average judge confidence
▲ with evidence per rule

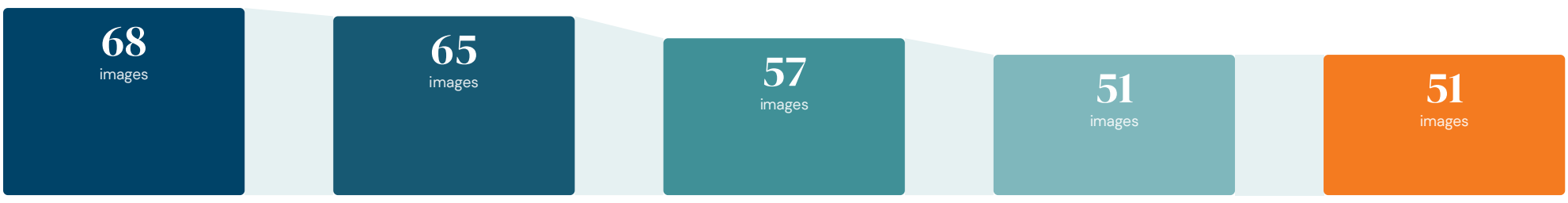


100 %
Image sets on the latest rulebook
▲ same rules in every studio

Rings: share of review time removed; share of images stopped by the gatekeepers; judge confidence; one segment per category rulebook. Figures indicative; validated metrics available on request.

01 Cheap filters first, the paid model last

One studio and agency batch through the gatekeepers · sample batch from the pipeline demo



Submitted
raw images from studio and agency
50 SKUs ingested · 38 routed to a rulebook, 12 to backlog

Tier 0 · decode
file opens and is an image
- 3 stopped
3 unreadable files

Tier 1 · basic
≥ 2000 × 2667 px, ≤ 3 MB
- 8 stopped
5 low-res · 3 over 3 MB

Tier 1.5 · ML
apparel classifier, right category
- 6 stopped
3 not apparel · 2 wrong category
1 blank frame

Tier 2 · VLM judge
clean sets only, rule by rule
17 of 68 never reach it
so they cost no paid VLM call

02 One image set, judged rule by rule

Boys graphic crew tee · Tops rulebook v1.3 · product images are drawn illustrations



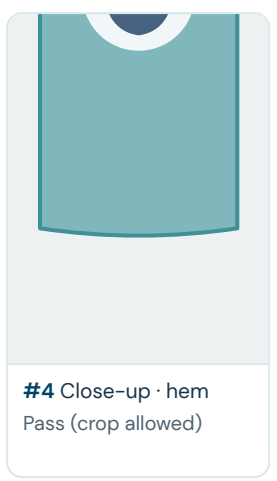
#1 Flat lay · front
Cropped right edge, off-centre



#2 Flat lay · back
Visible creases



#3 Close-up · collar
Pass (crop allowed)



#4 Close-up · hem
Pass (crop allowed)

A Cropping: right edge cut off by the frame **B** Creases: steam and reshoot

Orange boxes = crop coordinates returned by the judge

Tier 2 VLM judge · QC report for this set

Each rule returns a score, an observation and the images it looked at

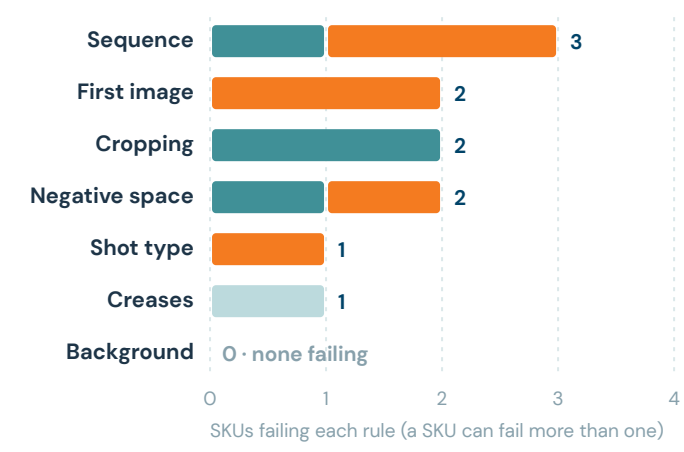
Sequence check	4/4 in the right position	PASS
Shot type check	4/4 of an allowed type	PASS
First image rule	#1 not fully visible and centred	FAIL
Cropping check	1/2 full shots uncropped	FAIL
Creases check	3/4 clear of creases	FAIL
Negative space check	2/2 full shots centred	PASS
Background check	4/4 on neutral grey	PASS

Verdict: reshoot. The set fails 3 of 7 checks: first image, cropping, creases. Image 1 is the right shot but is not fully visible and centred. Photographer gets the note while the set is still up; QA makes the final call.

03 From style guide to reshoot list

Rule registry and the batch QC report · sample data from the pipeline demo

SKUS JUDGED	PASS	TO RESHOOT
38	74%	10



SKUs failing each rule (a SKU can fail more than one)

Studio A	Studio B	Agency X
3 of 13 to reshoot	1 of 13 to reshoot	6 of 12 to reshoot

Rule registry: one JSON rulebook per category

Required image sequence (dark = full shot, no cropping allowed; light = close-up) · versioned by season

CATEGORY	IMAGES	SEQUENCE	VERSION	STATUS
Tops	4	■■□□	v1.3	Approved
Dresses	5	■■□□□	v1.1	Approved
Top & bottom sets	7	■■■■■□□	v1.0	AI draft
Outerwear & gilets	4	■■□□	v1.2	Approved
Bottoms	4	■■□□	v1.0	Approved
Nightwear sets	5	■■■□□	v0.9	AI draft

```
{ "category": "Tops", "rulebook_version": "v1.3", "required_images": 4,
  "sequence": [ { "position": 1, "shot_type": "flat-front", "crop_allowed": false }, ... ],
  "quality_standards": { "background": "Neutral light grey (#EDEDED) or white seamless", ... }
```

04 Four pain points, four fixes

Before → after

BEFORE
Massive style guides
Sequence, shot types, framing, creases and backgrounds differ by category and season; shared as PDFs and screenshots.

GUIDELINE SCRAPER + VLM

AFTER
JSON rulebooks
Scraped and turned into versioned, machine-readable rules per category.

BEFORE
Thousands of raw images
Every season, from several studios and agencies; junk reviewed like everything else.

TIER 0 · 1 · 1.5

AFTER
Junk stopped cheaply
Decode, resolution, file-size and classifier checks run before the VLM.

BEFORE
Hours of manual QA
Each image compared to the guide by eye, set by set; hours per batch.

TIER 2 VLM JUDGE

AFTER
Rule-by-rule verdicts
Scores, evidence and crop coordinates; QA reviews the report and makes the final call.

BEFORE
Subjective and late
Different reviewers, different answers; reshoots found after the set was struck.

QC WEB APP

AFTER
Instant, consistent
Same rules in every studio; feedback while the photographer is on set.

05 Where QA time goes

Before vs after · indicative

Before 48 h



After 5.1 h



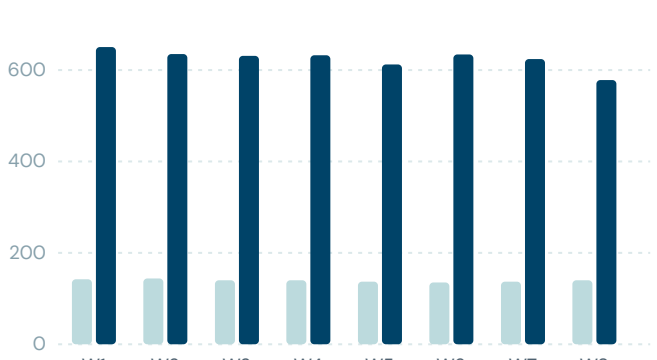
3.2 h of it reviewing the AI report

- Checking images by eye 38 → 0
- Rework emails 5 → 0.3
- Reshoot coordination 5 → 1.3
- Reviewing the AI report 0 → 3.2
- Rulebook upkeep 0 → 0.3

Hours per 1,000 images, by activity

06 SKUs cleared per day

Weekday average by week, before (light) and after (navy) · indicative



139 → **625** avg SKUs per weekday

07 How HMC ran it

2 weeks	**Discover** Sat with QA and photographers; rejection taxonomy and pipeline design.
2 weeks	**Design** Guidelines as JSON: rulebook schema and rule registry.
6 weeks	**Build** Scraper, router, Tier 0 / 1 / 1.5 gatekeepers and Tier 2 judge in one app.
1 studio	**Adopt** Kidswear studio pilot; photographers see results first.
All categories	**Scale** Every category and studio; agency uploads checked; feeds catalogue go-live.

What any e-commerce studio can borrow

01
Turn the guide into data first
A JSON rulebook is something every studio, agency and model can follow the same way.

02
Filter cheaply, judge carefully
Basic checks and small classifiers stop junk for a fraction of the cost of a vision model.

03
Explain every verdict
A score, an observation and the exact image and crop make the AI's call easy to trust and to act on.

Built with **Azure OpenAI GPT-4.1 mini** · **Guideline scraper** · **JSON rule registry** · **ML classifiers** · **Python** · **QC web app** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

HMC ONE
AI STRATEGY. BUILT TO SCALE.